

Smart Snow Goggle for Winter Sports

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Background and Motivation

Smart snow goggles are a wearable display concept for winter sports, giving users hands-free access to information like speed, altitude, and temperature. Unlike current products that can be heavy and limited, they aim to offer a lighter, more practical, and more interactive solution for skiers and snowboarders.

Project Goal

Develop a smart snow goggle device that provides winter sport participants with real-time environmental data to enhance situational awareness.

Key Requirements & Tested Results

Table 1: Key Requirements & Test Results

Requirements	Results
Speed accuracy: ± 0.5 km/h	Speed error: 0.47 km/h average absolute error
Temperature accuracy: ± 0.5 °C	Temperature error: 0.50 °C average absolute error
Altitude accuracy: ± 10 m	Altitude error: 27.25 m
Media capture: 1080p @ 30 FPS	Photo resolution up to 4608×2592 , Video resolution up to 1080p @ 30 fps
Mass: < 500 g	Final mass: 614 g
Dimensions: $\leq 16 \times 8 \times 5$ cm	Dimensions: $19.8 \times 17.5 \times 8.6$ cm
Battery life: ≥ 2 hours	Battery life: ≥ 2.5 hours

Design Overview

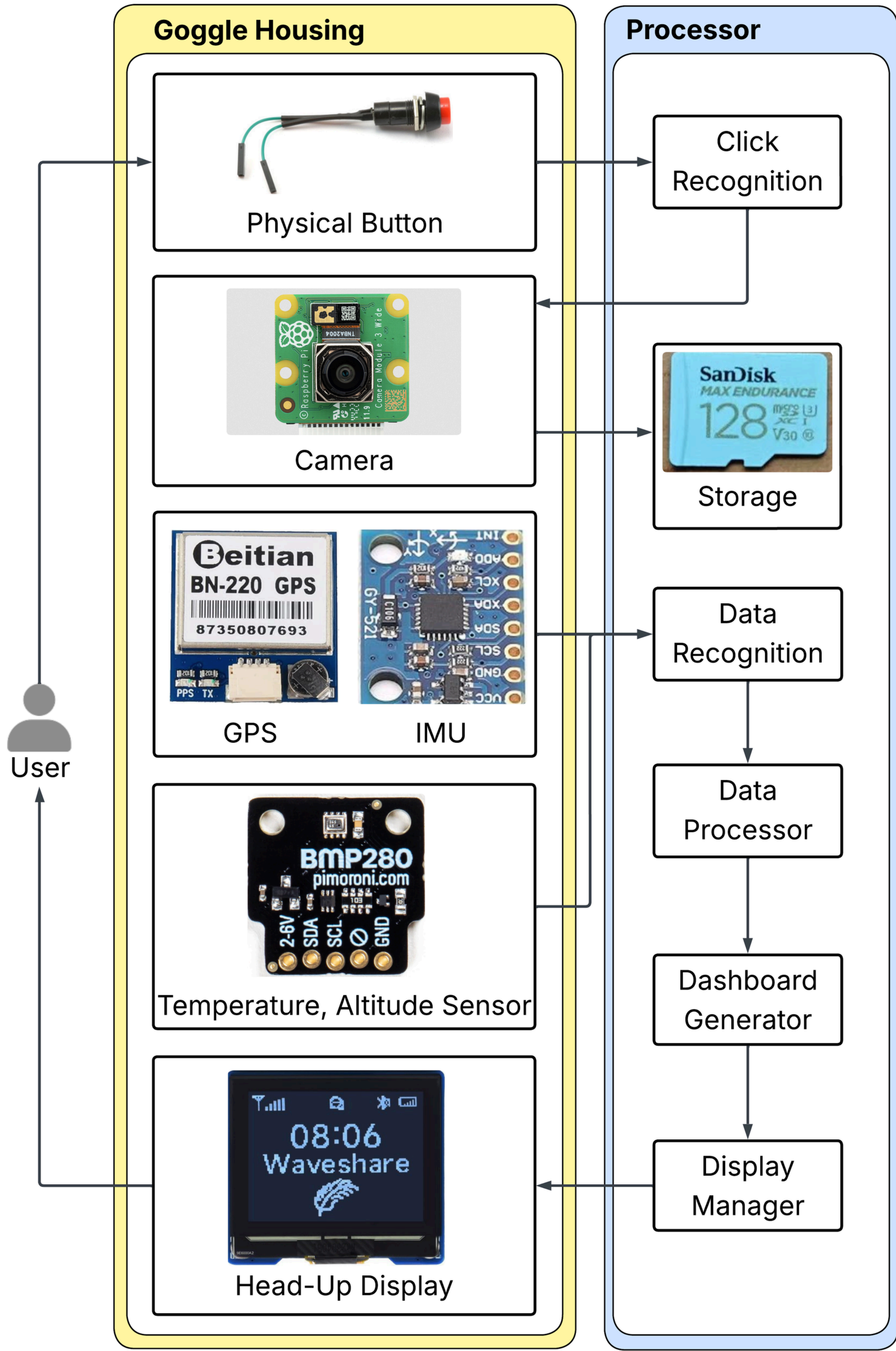


Figure 1: System Block Diagram with module demonstrations

Conclusion

- The system achieved reliable speed, temperature, and video performance.
- Current limitations include altitude accuracy and oversized packaging.
- Require additional field tests to improve usability and robustness.

Challenges

Table 2: Challenges

Aspect	Description
Spatial Constraints	Required specific distance for clear HUD reflection, resulting in a thick housing.
Packaging Efficiency	Prototype exceeded size limits, weighing 614 g and measuring $19.8 \times 17.5 \times 8.6$ cm due to large battery.
Signal Acquisition	GPS module struggled with satellite signals indoors, hindering development and testing.

Highlights of Accomplishments

- Functional proof-of-concept: built a working snow goggle integrating sensing, processing, media capture, storage, and HUD visualization.
- Successful hardware integration: custom goggle housing accommodated Raspberry Pi, camera, button, GPS, IMU, BMP280 sensor, and HUD.

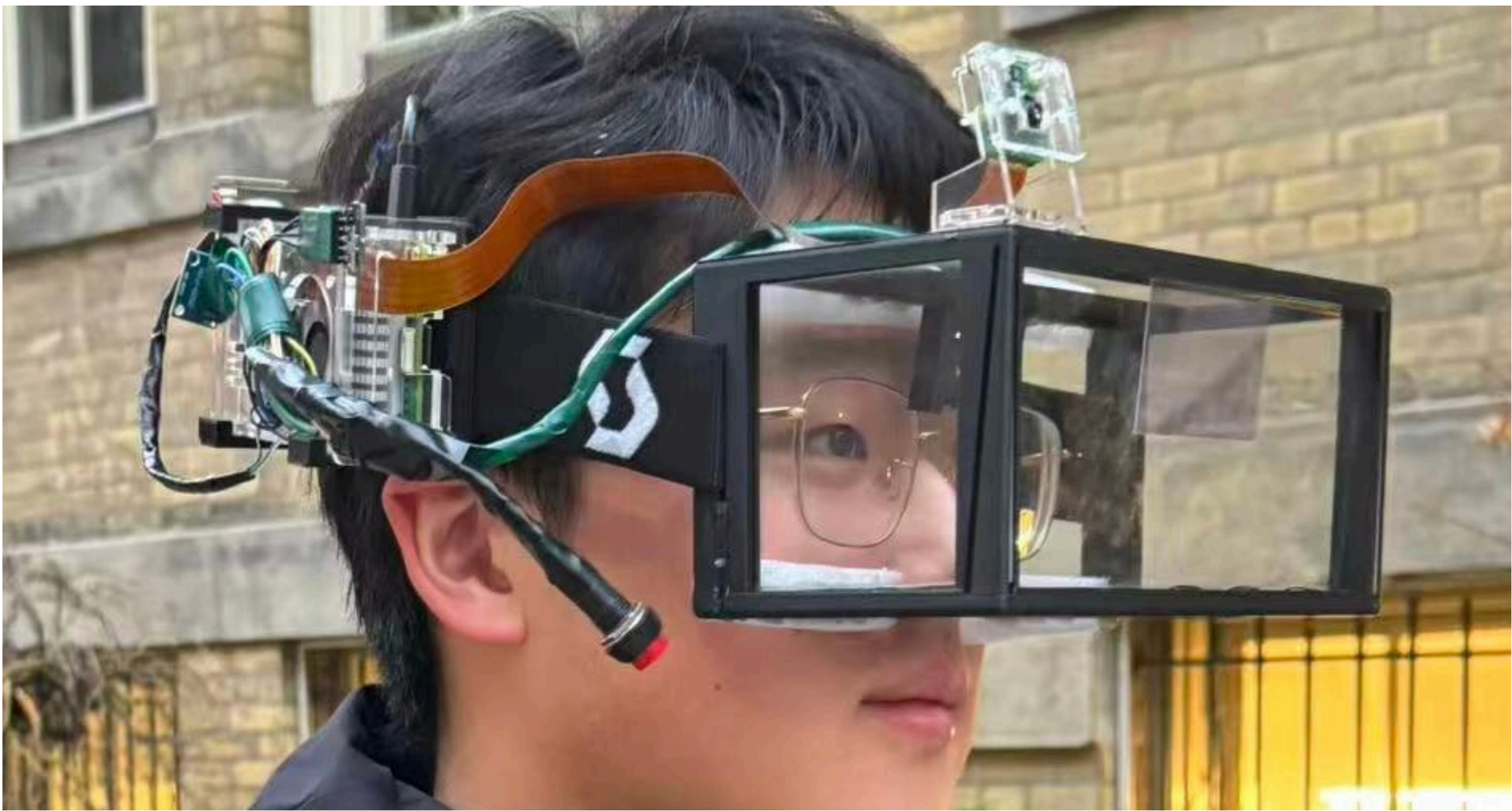


Figure 2: Use case example of the final design

Future Works

- Improve altitude accuracy through better sensor calibration and fusion of GPS, IMU, and pressure data.
- Reduce device size, weight, and power consumption through PCB integration and optimized packaging.
- Enhance user experience with improved display readability, smarter interactive features, and more field testing in real winter conditions.